Changing the temperature boundaries

Up to now we've been almost completely focused on the heat equation with two specific spatial conditions and one initial temperature condition,

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$$

with $u(0,t) = 0$ and $u(L,t) = 0$

and $u(x,0) = f(x)$

These two boundary conditions define the temperature of the material as $0°$ at $x = 0$, and $0°$ at $x = L$. In other words, we've been modeling a one-dimensional temperature distribution, in which temperature only flows left and right, and the temperature at both ends of the system is $0°$.

The system doesn't "leak" heat in any other direction as it flows from one end of the material to the other, or vice versa, because the system is perfectly insulated. And there are no other heat inputs to the system.

While defining the temperature as $0°$ at both ends of this kind of one-dimensional system makes it easier to solve the heat equation, doing so also makes the solution less useful.

If at all possible, we'd prefer to be able to choose some non-zero values for these temperatures, and still be able to solve the heat equation.

Non-zero temperature boundaries

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Instead of defining the temperature at the ends of the system as $u(0,t) = 0$ and $u(L,t) = 0$, let's choose two non-zero temperatures, and rewrite our boundary value problem as

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$$

with $u(0,t) = T_1$ and $u(L,t) = T_2$

and $u(x,0) = f(x)$

As it turns out, this small change in the boundary conditions means we can no longer use the product solution we've been using up to this point, $u(x,t) = v(x)w(t)$. Instead, if we assume that we won't add heat into the system, and that the system won't leak heat, and therefore that temperature changes can only occur at the ends of the system, then we know that the material will eventually reach an equilibrium temperature, if we take time t out far enough.

$$\lim_{t\to\infty} u(x,t) = u_E(x)$$

The function $u_E(x)$ models the equilibrium temperature, and it'll satisfy

$$\frac{d^2u_E}{dx^2} = 0 \qquad u_E(0) = T_1 \qquad u_E(L) = T_2$$

The characteristic equation associated with this second order equation, $u_E'' = 0$, is

$$r^2 = 0$$

$$r = 0,\ 0$$

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Because we get equal real roots, $r_1 = r_2 = 0$, the general solution is

$$u_E(x) = c_1 e^{r_1 x} + c_2 x e^{r_1 x}$$

$$u_E(x) = c_1 e^{0x} + c_2 x e^{0x}$$

$$u_E(x) = c_1 + c_2 x$$

If we substitute the boundary condition $u_E(0) = T_1$, we get

$$T_1 = c_1 + c_2(0)$$

$$c_1 = T_1$$

If we then substitute $c_1 = T_1$ and $u_E(L) = T_2$, we get

$$T_2 = T_1 + c_2 L$$

$$c_2 = \frac{T_2 - T_1}{L}$$

So **equilibrium temperature** is modeled by

$$u_E(x) = T_1 + \frac{T_2 - T_1}{L} x$$

With this value of $u_E(x)$, it can be shown that the solution to the original differential equation is

$$u(x, t) = u_E(x) + v(x, t)$$

where

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$$v(x,t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-k\left(\frac{n\pi}{L}\right)^2 t}$$

$$B_n = \frac{2}{L}\int_0^L (f(x) - u_E(x))\sin\left(\frac{n\pi x}{L}\right)\, dx \qquad n = 1,\ 2,\ 3,\ \ldots$$

Which means that the solution to the heat equation with non-zero temperature boundaries is

$$u(x,t) = u_E(x) + v(x,t)$$

$$u(x,t) = T_1 + \frac{T_2 - T_1}{L}x + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-k\left(\frac{n\pi}{L}\right)^2 t}$$

$$B_n = \frac{2}{L}\int_0^L (f(x) - u_E(x))\sin\left(\frac{n\pi x}{L}\right)\, dx \qquad n = 1,\ 2,\ 3,\ \ldots$$

Let’s do another example, this time where we solve the heat equation with different boundary conditions.

Example

Given a one-dimensional temperature distribution, if temperature at $x = 0$ is $10°$, temperature at $x = L$ is $20°$, and $u(x,0) = 40$, find an equation that models heat in the system.

We’re solving the heat equation with two boundary conditions and an initial condition.

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$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$$

with $u(0,t) = 10$ and $u(L,t) = 20$

$$u(x,0) = 40$$

First, we'll find the function modeling equilibrium temperature.

$$u_E(x) = T_1 + \frac{T_2 - T_1}{L}x$$

$$u_E(x) = 10 + \frac{20 - 10}{L}x$$

$$u_E(x) = 10 + \frac{10}{L}x$$

Next, we'll find the coefficients B_n.

$$B_n = \frac{2}{L}\int_0^L (f(x) - u_E(x))\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2}{L}\int_0^L \left(40 - \left(10 + \frac{10}{L}x\right)\right)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2}{L}\int_0^L \left(40 - 10 - \frac{10}{L}x\right)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{20}{L}\int_0^L \left(3 - \frac{1}{L}x\right)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

Split the integral in two,

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$$B_n = \frac{60}{L}\int_0^L \sin\left(\frac{n\pi x}{L}\right)\,dx - \frac{20}{L^2}\int_0^L x\sin\left(\frac{n\pi x}{L}\right)\,dx$$

then integrate, using integration by parts for the second integral with $u = x$, $du = dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$B_n = -\frac{60}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_0^L - \frac{20}{L^2}\left[-\frac{Lx}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \frac{L}{n\pi}\int\cos\left(\frac{n\pi x}{L}\right)\,dx\right]\Big|_0^L$$

$$B_n = -\frac{60}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_0^L - \left[\frac{20}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right) - \frac{20x}{n\pi L}\cos\left(\frac{n\pi x}{L}\right)\right]\Big|_0^L$$

Evaluate over the interval.

$$B_n = -\frac{60}{n\pi}\cos(n\pi) + \frac{60}{n\pi}\cos(0)$$

$$-\left(\frac{20}{(n\pi)^2}\sin(n\pi) - \frac{20}{n\pi}\cos(n\pi) - \frac{20}{(n\pi)^2}\sin(0) + \frac{20(0)}{n\pi L}\cos(0)\right)$$

$$B_n = -\frac{60}{n\pi}\cos(n\pi) + \frac{60}{n\pi}\cos(0) - \frac{20}{(n\pi)^2}\sin(n\pi)$$

$$+\frac{20}{n\pi}\cos(n\pi) + \frac{20}{(n\pi)^2}\sin(0) - \frac{20(0)}{n\pi L}\cos(0)$$

$$B_n = -\frac{60}{n\pi}(-1)^n + \frac{60}{n\pi} + \frac{20}{n\pi}(-1)^n$$

$$B_n = \frac{60}{n\pi} - \frac{40}{n\pi}(-1)^n$$

Then the solution to this heat equation is

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$$u(x,t) = T_1 + \frac{T_2 - T_1}{L}x + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-k\left(\frac{n\pi}{L}\right)^2 t}$$

$$u(x,t) = 10 + \frac{10}{L}x + \sum_{n=1}^{\infty} \left(\frac{60}{n\pi} - \frac{40}{n\pi}(-1)^n\right) \sin\left(\frac{n\pi x}{L}\right) e^{-k\left(\frac{n\pi}{L}\right)^2 t}$$

$$u(x,t) = 10 + \frac{10x}{L} + \frac{20}{\pi} \sum_{n=1}^{\infty} \frac{3 + 2(-1)^{n+1}}{n} \sin\left(\frac{n\pi x}{L}\right) e^{-k\left(\frac{n\pi}{L}\right)^2 t}$$

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